📦 Dependencies

🧰 Software Libraries

DHT Sensor Library

Used to read temperature data from the DHT11 sensor.

Adafruit Unified Sensor Library

Required dependency for the DHT library to function properly.

💻 Development Environment

Arduino IDE

For writing and uploading the code to the Arduino board.

🔌 Hardware Components

Arduino Board (e.g., Arduino Uno)

Soil Moisture Sensor

DHT11 Temperature Sensor

Relay Module (Active LOW)

Water Pump

Jumper Wires

External Power Supply (for the pump)

⚙️ Installation

Open Arduino IDE

Go to Library Manager

Install:

DHT sensor library by Adafruit

Adafruit Unified Sensor

⚠️ Notes

Make sure the relay type (Active LOW / Active HIGH) matches your code.

The pump must NOT be powered directly from Arduino.

Ensure proper wiring to avoid damaging components. 🌱 Smart Irrigation System

📌 Description

A smart Arduino-based irrigation system designed to automate plant watering based on soil moisture levels. The system uses a soil moisture sensor and a water pump to

ensure plants receive the right amount of water without manual intervention, improving water efficiency and plant health. It also monitors ambient temperature for environmental awareness.

⚙️ How It Works

The system continuously reads data from the soil moisture sensor.

If the soil is dry, the system automatically turns the pump ON to water the plant.

If the soil is wet, the pump is turned OFF to prevent overwatering.

Temperature readings are displayed for monitoring purposes.

🚀 How to Use

Hardware Setup

Connect the soil moisture sensor to the Arduino.

Connect the DHT11 temperature sensor.

Connect the relay module to control the water pump.

Power the pump using an external power source (through the relay).

Upload the Code

Open Arduino IDE.

Upload the project code to your Arduino board.

Run the System

Open the Serial Monitor to view temperature and soil status.

Place the soil sensor in the plant’s soil.

The system will automatically control watering based on soil condition.